GOC

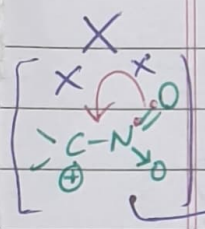
1. +I effect (eRG) $-O^- > -\overset{\overset{O}{\|}}{C}-O^- > R_3C- > R_2CH- > RCH_2 > D$

   −I effect (eWG) $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{S}R_2 > -\overset{\oplus}{N}H_3 > NO_2 > SO_2R > CN > SO_2Ar >$
   $-COOH > F > Cl > Br > I > -O-Ph > -\overset{\overset{O}{\|}}{C}-OPh > -OR > -\overset{\overset{O}{\|}}{C}-R >$
   $-SH > -SR > -OH > -C\equiv C-R' > -Ph > -CH=CR_2$

2. Resonance. { eg.:- $-C=C-C=C-$ $\quad -C=C-O^{\ominus}$ $\quad -C=C-\dot{C}$ }

   mesomeric → a type of resonance.

   net
   ↳ hybrid ⇒ all structures may have diff contribution → Stability → 1) no. of bonds
   2) -ve on eN↑ atom
   3) charged/uncharged

   ↳ (R) eN atom se ulta nahi hota!

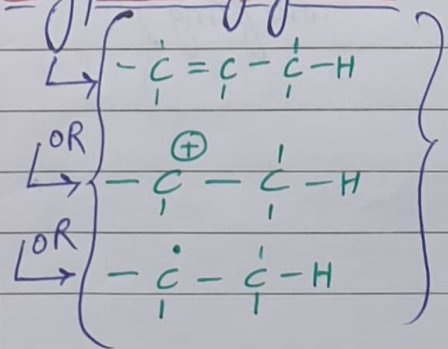
3. e^- deficient ⇒ E^+ electrophile — e^- rich ⇒ Nu^-

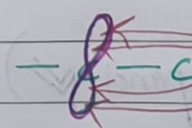
   $H^+, AlCl_3, BF_3, Cl^+, Br^+$ — $\ddot{N}H_3, H_2\ddot{O}:, Cl^-, Br^-$ nucleophile.

   NH_4 / H_3O^+ etc. ⇒ xxx E^+ nor Nu^- xxx

   ★ Nu^- xxx attack $-C=C-$ (repel maarega!)
   but can attack $-\overset{\delta+}{C}=O$

4. Hyperconjugation ⇒ σ − π overlap!

   ★ $H_2C^{\oplus}-C(H)=CH_2$ → zero HC for C^+!

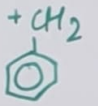
   (H) ← out of plane of π

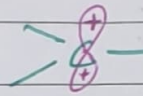
   no Hyperconjugation xxx.

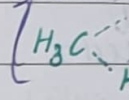
$CH_2=\overset{\oplus}{C}H$ → vinyl

 → phenyl C^+

$CH_2=CH-\overset{+}{C}H_2$ → Allylic

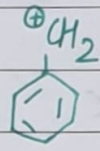
 → benzylic

5. Carbocation  sp^2 (AKA carbenium)

{carbonium is  }

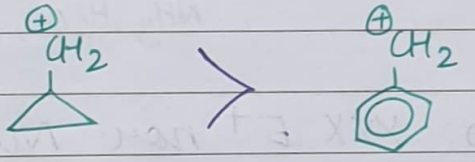
↳ Stability ⇒ (I) Aromaticity
(II) Resonance
(III) Hyperconjugation
(IV) Inductive.

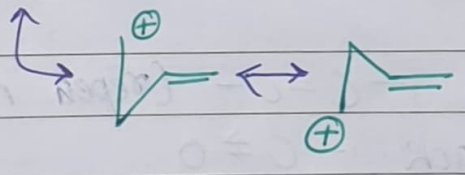
Typical cases:

a) $CH_3-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C^+}}}}$ >  > $CH_3-\overset{\oplus}{C}H-CH_3$ ≥ $CH_2=CH-\overset{\oplus}{C}H_2$

b) $-\overset{\oplus}{\underset{|}{C}}-F$ > $-\overset{\oplus}{\underset{|}{C}}-Cl$ > $-\overset{\oplus}{\underset{|}{C}}-Br$ > $-\overset{\oplus}{\underset{|}{C}}-I$

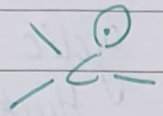
{ $-\underset{|}{C}$ > F/Cl/Br/I (−I) lekin → $-\overset{\oplus}{\underset{|}{C}}-\ddot{X}$ (+R) agar C^+ bann gya }

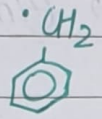
c) 

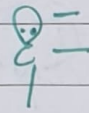


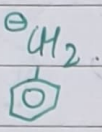
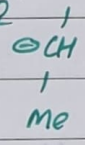
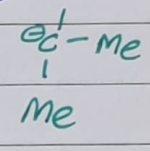
★ Rearrangement ⇒ C^+ ko zyada stable banane ke liye!
(stability must increase while shifting).

(ring expand ✓✓)

6. **Free Radical**  sp^2

Stability ⇒  > $CH_2=CH-\dot{C}H_2$ > $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\dot{C}}}$ > $CH_3-\dot{C}H-CH_3$ > $CH_3\dot{C}H_2$

7. **Carbanion**  sp^3

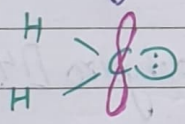
Stability ⇒  > $CH_2=CH-\overset{\ominus}{C}H_2$ > $\overset{\ominus}{C}H_3$ > $CH_3\overset{\ominus}{C}H_2$ >  > 

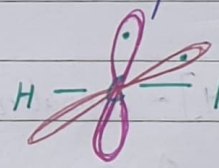
8. Carbene : CH_2

Spin multiplicity $= 2|S|+1$

Singlet $\frac{1}{2}-\frac{1}{2}=0$ $\therefore 2(0)+1 = \boxed{1}$

Triplet $\frac{1}{2}+\frac{1}{2}=1$ $2(1)+1 = \boxed{3}$

Singlet sp^2 

Triplet sp 

generally, Stability $<$ (∵ pairing energy in singlet)

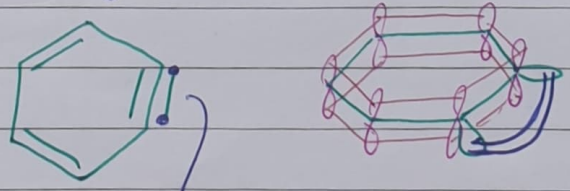
9. Back Bonding (−I)

$:CF_2 > :CCl_2 > :CBr_2 > :CI_2$ — Singlet

$:CF_2 < :CCl_2 < :CBr_2 < :CI_2$ — Triplet.

10. **Benzyne** (dehydrobenzene)

Aromatic in nature.



→ sp^2-sp^2 overlap!

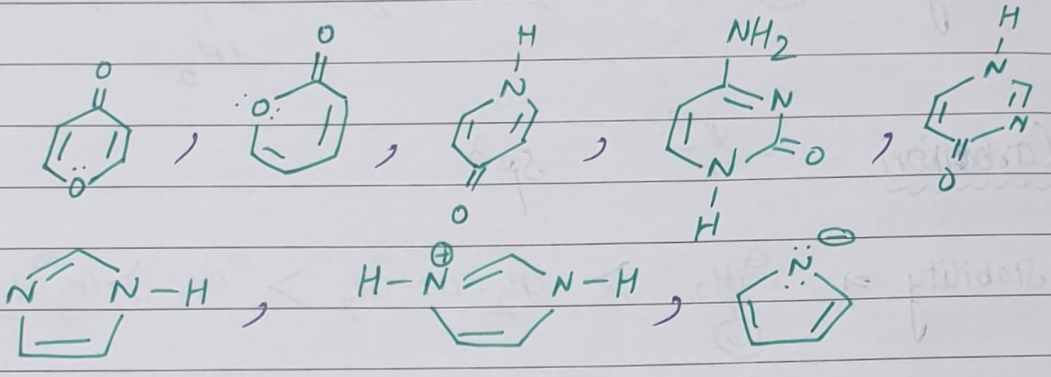
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all sp^2 atoms!

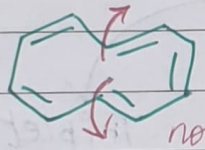
11. Aromaticity ⟶ 1. cyclic + conjugated + Planar
2. Huckle rule
↳ $(4n+2)$ π e⁻s. ⇒ Aromatic.

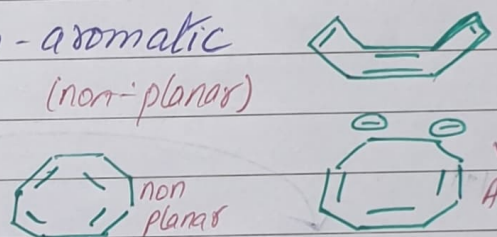
if 4n π e⁻s ⇒ anti Aromatic

typical eg:-



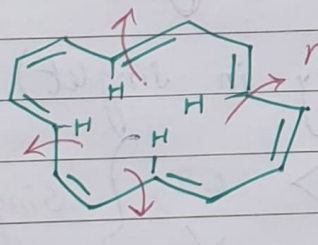
★ [8]- Annulene is non-aromatic
[10]- Annulene " (non-planar)

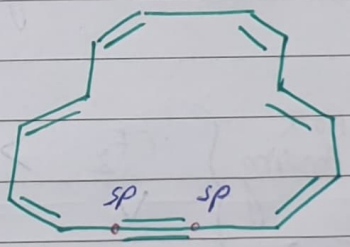




cyclooctatetrene — cyclooctatetrene dianion

[14]- Annulene non-aromatic BUT dehydro [14]Annulene aromatic ✓





(sirf ek π bond count hoga conjugation mein)